

Revised Generation Interconnection System Impact Study Report

For Queue Project Q11-0029

Battle Creek 100 kV
March 18, 2013

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q11-0029. The Interconnection Customer proposes a 150.9 MW wind facility interconnecting at the Battle Creek 100 kV substation in the Illinois area. The request was received on April 20, 2011. The study evaluated system intact and single contingency conditions using the regional planning case, including 22 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 4 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$16,467,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

The Battl Creek-Range Line Sw 100 KV is overloaded to 142.7% of its 106 MVA emergency rating for the loss of the Sterling Steel-Battle Creek 100 KV line. The project contributes 100.0% DFAX (150.9 MW) to this overload.

Battle Crk-Sterling Steel 100 KV: 142.7% loading (114 MVA) following the outage of the George Rd-Jackson-Sunrise 100 KV; DFAX = 73.1%.

The Blountville Sw Station Substation-Jackson-Sunrise 100 KV line is overloaded to 112.9% of its 98 MVA emergency rating for the loss of the Keahole Substation-Sterling Steel 100 KV. The project contributes 72.0% DFAX (108.7 MW) to this overload.

Jackson-Sunrise-George Road 100 KV: 105.6% loading (151 MVA) following the outage of the Battle Crk-Sterling Steel 100 KV; DFAX = 54.7%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Battl Creek-Range Line Sw 100 KV	Rebuild line with higher-capacity conductor	\$5,563,000	100% (\$5,563,000)
Battle Creek-Sterling Steel 100 KV	Rebuild line with higher-capacity conductor	\$911,000	100% (\$911,000)
Jackson-Sunrise-Blountville Sw Sta 100 KV Line	Rebuild line with higher-capacity conductor	\$4,654,000	100% (\$4,654,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Jackson-Sunrise-George Road 100 KV	Reconductor line	\$5,339,000	100% (\$5,339,000)
	Total allocated to project		\$16,467,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.